TorusFlow: Topological Phase Reconstruction from Spatiotemporal Data

Problem statement

Given synchronized sensor readings $Y = [y_0, \dots, y_{N-1}] \in \mathbb{R}^{m\times N}$ sampled every Δt, recover their independent phases. Assume

$$y_n = g(\theta_n) + \varepsilon_n, \qquad \theta_{n+1} = \theta_n + \alpha \pmod{2\pi}, \qquad \theta_n \in \mathbb{T}^r = (\mathbb{R}/2\pi\mathbb{Z})^r.$$

The phase count r, smooth observation map $g : \mathbb{T}^r \to \mathbb{R}^m$ and phase increment $\alpha \in \mathbb{R}^r$ are unknown; $\varepsilon_n \in \mathbb{R}^m$ is noise. Assume $k^\top \alpha \notin 2\pi\mathbb{Z}$ for nonzero $k \in \mathbb{Z}^r$, so the ideal orbit explores the full torus.

For $1 < L < N$, form SSA/MSSA windows [1, 2]:

$$x_n = (y_n^\top, \dots, y_{n+L-1}^\top)^\top \in \mathbb{R}^{mL}, \qquad H = [x_0, \dots, x_{K-1}], \quad K = N - L + 1.$$

Their noiseless map is $h_L(\theta) = (g(\theta)^\top, \dots, g(\theta + (L-1)\alpha)^\top)^\top$. Use x_n^{p} and h_L^{p} for windows after noise reduction, trend removal and scaling. Assume h_L^{p} is a C^2 embedding [3] and the data resolve the geometry and cycles of $M = h_L^{\mathrm{p}}(\mathbb{T}^r)$.

Objective. Estimate r, phases Φ, tangent fields E_j and an inverse window map Ψ from the readings:

$$\varphi_n = \Phi(x_n^{\mathrm{p}}) \in \mathbb{T}^r, \qquad \varphi_{n+1} \approx \varphi_n + \widehat{\alpha} \pmod{2\pi},$$
$$D\Phi_z E_j(z) \approx e_j, \qquad \Psi(\varphi_n) \approx x_n^{\mathrm{p}}.$$

Here $\widehat{\alpha} \in \mathbb{R}^r$ is constant, e_j is the jth coordinate vector and $E_j(z) \in \mathbb{R}^{mL}$ moves a window $z \in M$ along phase j. Phases are determined up to an origin and an integer coordinate change with determinant ± 1.

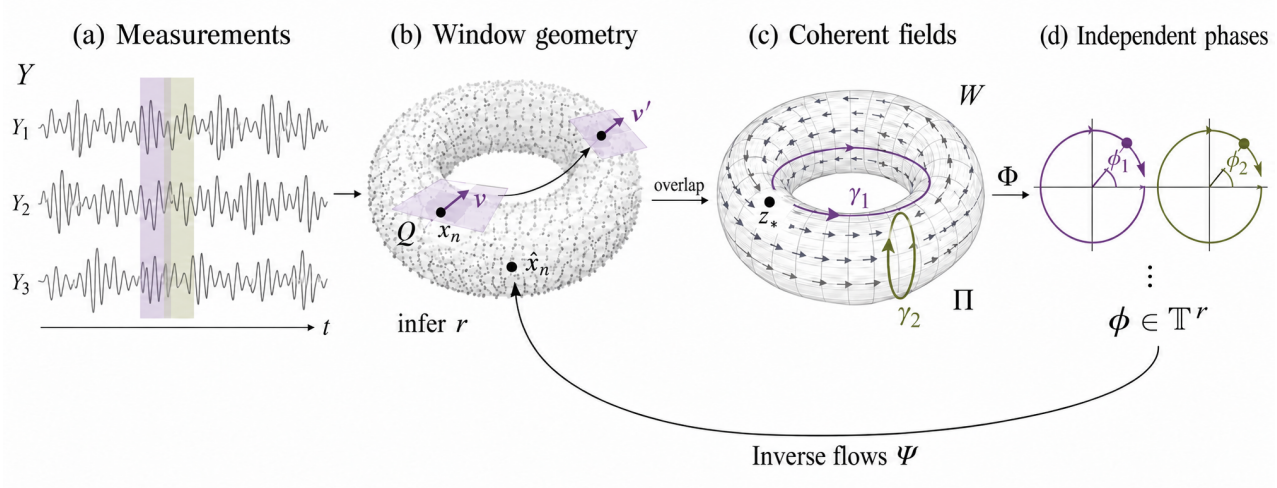


Figure 1: From sensor readings to independent phases and back. The illustrated torus has $r = 2$; the method estimates r from the windows.

1. Windows and tangent geometry

Each window describes several consecutive readings from all sensors. From the resulting point cloud, we estimate the number of phases and the local directions in which windows can move.

- **Prepare windows.** Denoise by MSSA [2], subtract a locally smoothed trend [4], then centre and scale each channel. Retain the trend for reconstruction. Stack L readings into x_n^{p}. Filter settings and candidate window lengths are listed in Appendix A.1.

- **Reduce coordinates.** SVD of the K_{p} prepared training windows gives

$$\widetilde{x}_n = U_D^\top x_n^{\mathrm{p}} \in \mathbb{R}^D, \qquad U_D \in \mathbb{R}^{mL\times D}, \quad U_D^\top U_D = I_D.$$

  Truncate at a clear singular-value gap only if the reconstruction error is small and the estimated local dimension is preserved. Otherwise retain the numerical span. Write $\widetilde{z} = U_D^\top z$.

- **Estimate r.** Centre neighbouring windows and apply SVD. A gap between large and small singular values proposes r. Under the torus model, each independent phase also contributes one full-turn cycle. We therefore test this count on the whole cloud:

  1. Starting with the first window, repeatedly select the window farthest from the selected set. Call these representative windows $z_v \in \mathbb{R}^{mL}$ *landmarks*; use their reduced coordinates $\widetilde{z}_v = U_D^\top z_v$ to measure distances.
  2. Join landmarks at distance below t and fill every triangle whose three edges are present. Filling triangles removes small loops created by neighbour connections; their boundaries do not count as full phase turns.
  3. Count independent graph cycles and remove combinations of triangle boundaries, using matrix ranks. For example, the outline of two adjacent triangles is also filled: their shared edge cancels.

  Increase t and track the remaining cycles [5, 16]. Require an interval with exactly r cycles. Choose t within it so that balls of radius $t/2$ around the landmarks cover the training windows. If local SVD is inconclusive, use the group of cycles with clearly longer lifetimes than the rest. Refine the landmark set if needed; keep the selected windows, edges and triangles. D counts linear coordinates; r counts phases.

- **Fit tangent planes.** Fit quadratic constraints $g_a(\widetilde{z}) \approx 0$, validated on withheld windows. Tangent velocities satisfy

$$\nabla g_a(\widetilde{z})^\top \widetilde{v} \approx 0, \qquad \widetilde{v} \in \mathbb{R}^D.$$

  If the solution space is not stably r-dimensional, differentiate local cubic surface fits [6, 7].

This gives orthonormal bases $\widetilde{Q}(\widetilde{z}) \in \mathbb{R}^{D\times r}$ and $Q_n = U_D \widetilde{Q}(\widetilde{x}_n) \in \mathbb{R}^{mL\times r}$. Keep the fitted geometry fixed. The surface fits are specified in Appendix A.2.

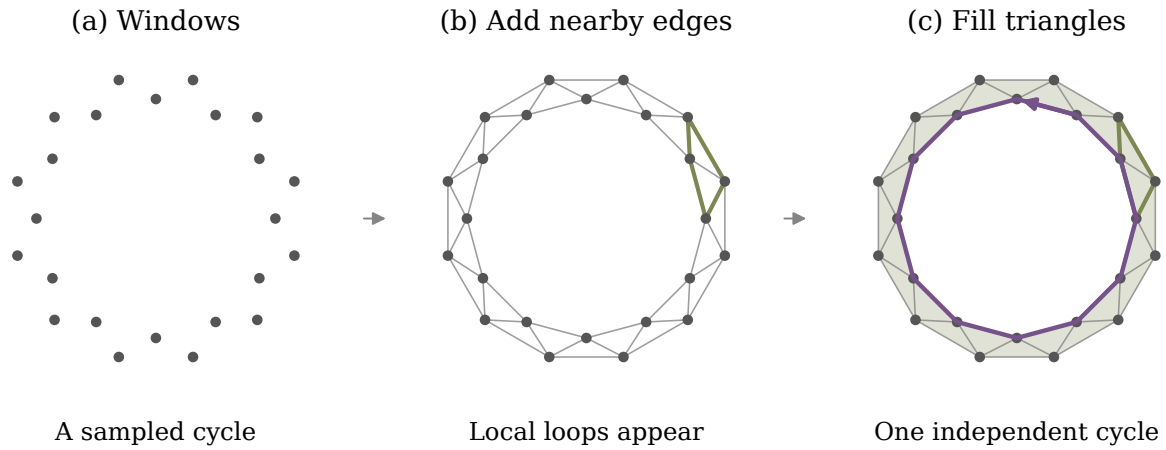


Figure 2: Why triangles matter. A schematic cloud around a circle. Neighbour connections create small loops (olive); filling their triangles leaves one independent cycle around the centre (purple). The graph is identical in (b) and (c).

2. Overlap transport and coherent fields

Local tangent bases can have different orientations at different windows. Matching changes in their shared readings aligns them into fields that represent the same phase changes along the record. Define

$$P_0 = \begin{bmatrix} I_{m(L-1)} & 0 \end{bmatrix}, \qquad P_1 = \begin{bmatrix} 0 & I_{m(L-1)} \end{bmatrix}, \qquad P_0, P_1 \in \mathbb{R}^{m(L-1)\times mL},$$

with m columns in each zero block. These matrices select the shared readings: $P_0 x_{n+1}^{\mathrm{p}} = P_1 x_n^{\mathrm{p}}$. Require rank $D(P_0 h_L^{\mathrm{p}}) = r$, so the overlap retains every tangent direction.

Transport. Match tangent changes on the shared readings:

$$A_n = \operatorname*{arg\,min}_{A\in\mathbb{R}^{r\times r}} \|P_0 Q_{n+1} A - P_1 Q_n\|_F^2.$$

Then $Q_n a$ is transported to $Q_{n+1} A_n a$ for $a \in \mathbb{R}^r$. Full column rank of $P_0 Q_{n+1}$ makes A_n unique.

Joint fit. Fit r tangent fields $W(z) \in \mathbb{R}^{mL\times r}$. The matrix R_n gives their coefficients in the local basis Q_n:

$$W_n := W(x_n^{\mathrm{p}}) = Q_n R_n, \qquad R_{n+1} \approx A_n R_n, \quad R_0 = I_r, \quad R_n \in \mathbb{R}^{r\times r}.$$

The initial basis prevents a zero solution. Blend local affine predictions and project onto the tangent planes. Fit their coefficients together [8], penalizing transport errors, disagreement at nearby windows and large coefficients. Freeze W, including its values between observations. Appendix A.3 gives the field model and fitting objective.

3. Coefficients of motion

Express motion as a combination of the fitted field directions. Integrating these coefficients along a path measures accumulated motion in one common basis. For a displacement $v \in \mathbb{R}^{mL}$ at z, solve for its frame coefficients $a = \beta_z(v) \in \mathbb{R}^r$:

$$(W(z)^\top W(z))a = W(z)^\top v.$$

For tangent v, this gives $v = W(z)a$. The ideal shift $T(h_L^{\mathrm{p}}(\theta)) = h_L^{\mathrm{p}}(\theta+\alpha)$ satisfies

$$DT_{h_L^{\mathrm{p}}(\theta)} Dh_L^{\mathrm{p}}(\theta) = Dh_L^{\mathrm{p}}(\theta+\alpha).$$

Since Dh_L^{p} spans the tangent space, exact transport preserves one invertible matrix $B \in \mathbb{R}^{r\times r}$ along the dense orbit; continuity gives

$$W(h_L^{\mathrm{p}}(\theta)) = Dh_L^{\mathrm{p}}(\theta) B, \qquad \beta_{h_L^{\mathrm{p}}(\theta)} Dh_L^{\mathrm{p}}(\theta) = B^{-1}.$$

Thus ideal cycle integrals measure full phase turns in the coordinates of W.

4. Full periods

A full turn returns to the same window but has a nonzero integral of β. The columns of Π record these integrals for r independent turns. Use the same landmarks, edges and triangles as in step 1.

1. **Identify full turns.** Assign integer edge labels $c_{uv} \in \mathbb{Z}^r$ using persistent cohomology [5]. Their sum along a closed path counts its signed turns. Filled triangles contribute zero:

   $$c_{vu} = -c_{uv}, \qquad c_{uu} = 0, \qquad c_{uv} + c_{vw} - c_{uw} = 0.$$

   Verify that every unit turn vector e_j occurs as a closed-path sum.

2. **Measure motion between landmarks.** For each landmark edge $u \to v$, find a shortest path in a nearest-neighbour graph of training windows, restricted to the covering balls. Its labels, added whenever the covering ball changes, must sum to c_{uv}. Compute $b_{uv} = \int_{u\to v} \beta \in \mathbb{R}^r$ using the fixed W.

3. **Fit the full periods.** Represent each integral as an endpoint difference plus full turns: $b_{uv} \approx f_v - f_u + \Pi c_{uv}$. The landmark coordinates $f_v \in \mathbb{R}^r$ and period matrix $\Pi \in \mathbb{R}^{r\times r}$ are unknown. Find them by least squares:

$$\min_{\{f_v\},\Pi} \sum_{u\to v} \|b_{uv} - (f_v - f_u + \Pi c_{uv})\|^2, \qquad f_0 = 0.$$

Along a closed path, the endpoint differences cancel. Thus the fitted integrals $\bar{b}_{uv} = f_v - f_u + \Pi c_{uv}$ sum to zero around a triangle and to $\Pi_{:,j}$ for one turn in direction j [10]. The field W remains fixed.

4. **Choose a shorter period basis.** Use the lattice-reduction step of toroidal coordinates (TC) [12] to choose shorter integer combinations of cycles. Measure their length through the average squared window motion:

$$\mathcal{G} = \Pi^\top \left(K_{\mathrm{p}}^{-1} \sum_n W_n^\top W_n \right) \Pi.$$

For a turn vector c, $c^\top \mathcal{G} c$ is the average of $\|W_n \Pi c\|^2$. For the resulting $U \in \mathbb{Z}^{r\times r}$ with $\det U = \pm 1$, replace $\Pi \leftarrow \Pi U$ and $c_{uv} \leftarrow U^{-1} c_{uv}$. This shortens the basis without changing the period lattice.

Appendix A.4 specifies the integer-label construction and the paths used for integration.

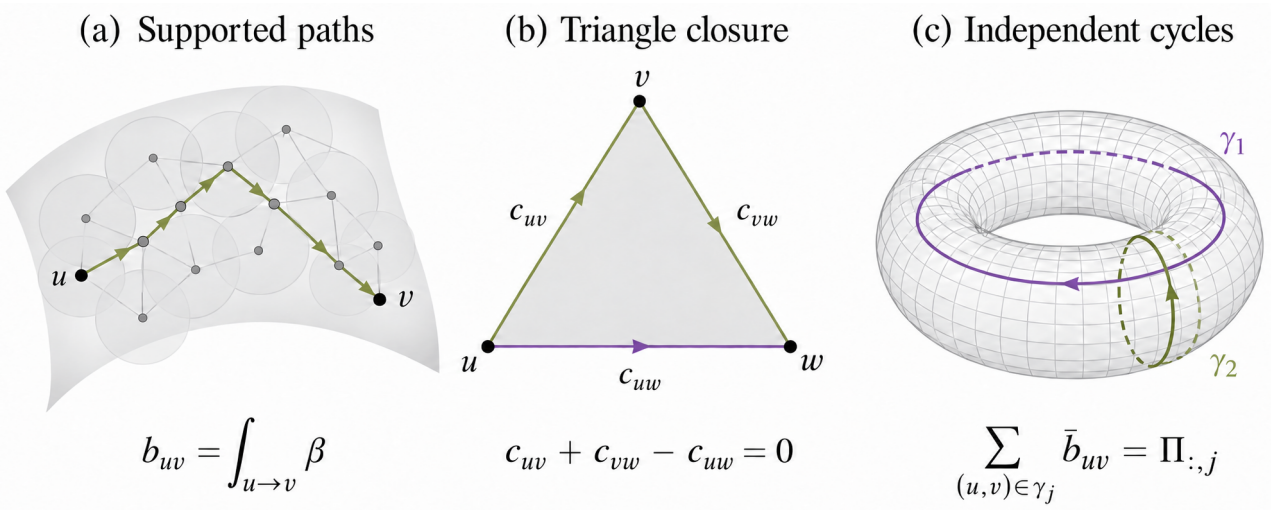


Figure 3: Paths, triangle closure and independent cycles. Paths respect the cover and edge labels; corrected cycle integrals give the columns of Π. The illustrated torus has $r = 2$.

5. Phase coordinates

The periods convert accumulated motion into angles. Assign values to the landmark windows and extend them to nearby windows using the fitted fields. At each z_v, set $q_v = \Pi^{-1} f_v$ (full turns). For a nearby window z, estimate

$$q_v + \Pi^{-1} \beta_{z_v} (z - z_v).$$

Local estimates may differ by whole turns, so align them before averaging. Add the integer turns c_{uv}, where u is any contributing landmark [11]:

$$\Phi(z) = 2\pi \sum_{v:\rho_v(z)>0} \rho_v(z) \left[q_v + c_{uv} + \Pi^{-1} \beta_{z_v} (z - z_v) \right] - \varphi_{\mathrm{ref}} \pmod{2\pi}.$$

The smooth weights $\rho_v \geq 0$ sum to one, vanish outside their landmark balls and reproduce each landmark value and derivative. Appendix A.5 (Phase interpolation) gives the weights as functions of distance and defines their values at landmarks. Contributing landmarks form filled triangles, so the angles are independent of u.

Phase origin. Initially take $z_* = x_0^{\mathrm{p}}$ as the reference window. The constant vector $\varphi_{\mathrm{ref}} \in \mathbb{R}^r$ is the value of the weighted sum, including 2π, at z_*. Subtracting it sets $\Phi(z_*) = 0$.

Use this fixed map for training and new windows; reject points outside the cover. It approximates the ideal integral

$$\Phi(z) = 2\pi\Pi^{-1} \int_{z_*}^{z} \beta \pmod{2\pi}.$$

6. Inverse reconstruction

Recover a window by starting at a reference window and changing each phase by its prescribed amount. Then combine overlapping windows to recover the sensor record. Ideally, $\beta W = I_r$ and $D\Phi = 2\pi\Pi^{-1}\beta$. To change only phase j at unit speed, solve

$$2\pi\Pi^{-1}a = e_j \quad \Longrightarrow \quad E_j(z) = W(z)a = \frac{1}{2\pi} W(z)\Pi e_j.$$

These fields commute in the exact model [13]. Starting at $u_0 = z_*$, reconstruct by successive flows [14, Theorem 9.46]:

$$\frac{dw_j}{ds} = E_j(w_j(s)), \qquad w_j(0) = u_{j-1}, \qquad u_j = w_j(\varphi_j), \qquad \Psi(\varphi) = u_r, \quad j = 1, \ldots, r.$$

Evaluate the fixed W along each flow. For local cubic geometry, failed training reconstruction or full-turn tests trigger normal corrections back to the same fitted surface. Rescale $\widehat{x}_n^{\mathrm{p}} = \Psi(\varphi_n)$, average copies of each sensor reading across overlapping windows [2], and restore the trend to obtain $\widehat{Y}$. Integration settings and acceptance thresholds are listed in Appendix A.6.

Algorithm 1 TorusFlow

Input: sensor readings Y, sampling interval Δt.
Output: r, Φ, E_j, Ψ, $\widehat{Y}$, or `unresolved`.
Skip candidates with unresolved geometry or deficient required ranks.

Denoise; subtract and retain the trend.
for each candidate window length and neighbourhood size **do**
  Scale signals; build windows; apply SVD; estimate r; fit tangent planes.
  Compute overlap transfers and jointly fit the fields W.
  Label cycles; integrate β along supported graph paths.
  Fit and reduce the period basis Π; construct Φ and E_j.
  Integrate E_j to obtain Ψ; average windows and restore the trend.
  return the fit if ranks, phase consistency and reconstruction pass.
  Otherwise continue with the next candidate.
end for; **return** `unresolved`.

Related methods

Table 1: Methods for phase reconstruction.

Method	Strengths	Limitations
SSA / MSSA [1, 2]	Separates oscillatory modes and reconstructs readings.	Linear rank counts harmonics as well as phases.
Circular coordinates (CC) [5]	Maps independent loops in a point cloud to angles.	Phase speed can vary; inversion needs another model.
Toroidal coordinates (TC) [12]	Reduces correlation between circular coordinates.	Phase speed can vary; inversion needs another model.
Koopman methods [13]	Finds dynamical phases and fields; supports reconstruction and forecasting.	Needs basis truncation, regularization and phase selection.
TorusFlow	Estimates r and phase fields; reconstructs readings using their flows.	Assumes observable torus dynamics; needs sample coverage and consistent fields.

Experiments

All fits use sensor readings only; the number of phases is estimated. We fit on the first part of each record and reconstruct a later observed block.

Signals and sensors

1. **Sine toy: one sensor, one phase.** The sensor records $y(t) = \sin t$ at $\Delta t = 0.4$, without noise or trend. This is a scalar record with no specified spatial position.

2. **Two-wave toy: twelve sensors, two phases.** Sensors at fixed positions $\xi_s \in [0,1]^2$ record two travelling waves:

$$y_s(n) = \cos(\theta_{n,1} - k_1^\top \xi_s) + 0.6\cos(\theta_{n,2} - k_2^\top \xi_s), \qquad k_1 = 2\pi(1, 0.5)^\top, \quad k_2 = 2\pi(-0.4, 1.1)^\top.$$

   We test clean readings and readings with a quadratic trend and 5% noise. The wave vectors k_1, k_2 give the propagation directions.

3. **Nonlinear signal: one sensor, one phase.** The sensor records

$$y(n) = \exp(0.4\cos\theta_n) + 0.25\sin\theta_n + 0.15\cos(2\theta_n),$$

   with a linear trend and 10% noise.

4. **Curved torus: four channels, three phases.** For phases (a, b, c), the readings are

$$y = \big((2+\cos c)\cos a,\ (2+\cos c)\sin a, (2+\sin c)\cos b,\ (2+\sin c)\sin b\big)^\top.$$

   The third phase changes the amplitudes of the first two. We test 0, 1, 5 and 10% noise, without a trend. These four channels are synthetic observables with no assigned spatial positions.

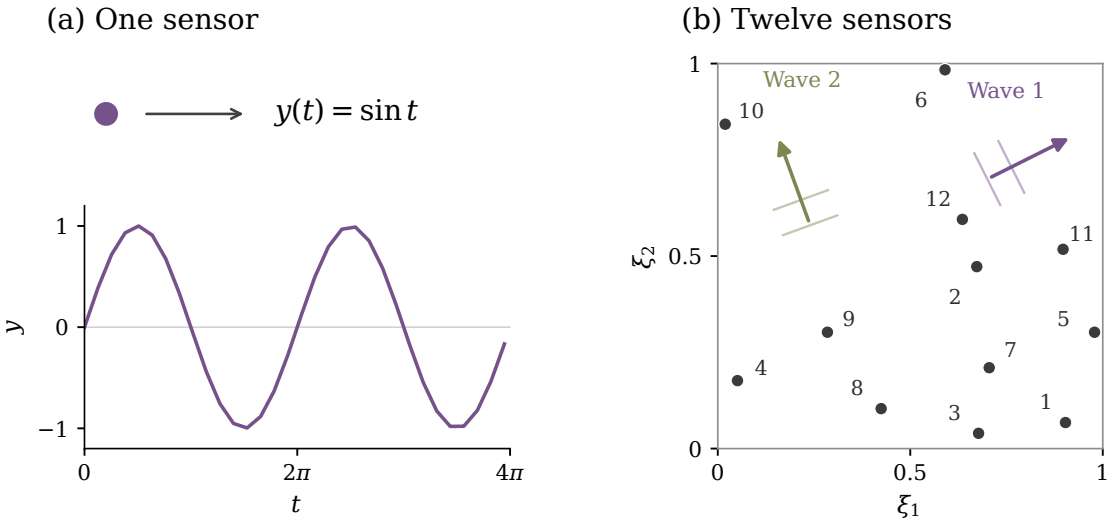


Figure 4: Toy measurements. Left: one recorded sinusoid. Right: the actual twelve sensor positions; arrows and wavefronts show how the two plane waves propagate.

Evaluation

We use two errors; smaller values are better.

- **Phase error (radians):** the mean angular distance to the reference,

$$E_\varphi = \mathrm{mean}_{n,j} \left| \mathrm{Arg}\, e^{i(\varphi_{n,j} - \theta^{\mathrm{ref}}_{n,j})} \right|.$$

  The reference θ^{ref} accounts for phase ordering, orientation and origin, fitted on training data only. $\mathrm{Arg} \in (-\pi, \pi]$.

- **Signal error (percent):** $100 E_Y$, where

$$E_Y = \sqrt{\mathrm{mean}_{n,s} \left(\frac{\widehat{Y}_{s,n} - Y^0_{s,n}}{\sigma_s} \right)^2}.$$

  Here Y^0 is the clean signal including the trend, and σ_s is the standard deviation of the clean oscillatory training signal at sensor s.

Phase errors use complete test windows; signal errors use retained test samples. Gaussian noise percentages are relative to the same σ_s. Appendix A.7 lists the data splits and scored samples.

Results

Each row is one realization; the four curved-torus runs use the same orbit and scaled noise.

Signal	Sensors	Noise (%)	$\widehat{r}$	Phase error (rad)	Signal error (%)
Sine toy	1	0	1	0.00020	0.496
Two-wave toy	12	0	2	0.00134	0.110
One nonlinear sensor	1	10	1	0.01157	2.499
Two waves	12	5	2	0.00431	0.695
Curved torus	4	0	3	0.00439	0.456
Curved torus	4	1	3	0.00933	0.650
Curved torus	4	5	3	0.01091	1.840
Curved torus	4	10	3	0.01393	2.285

Table 2: TorusFlow results. $\widehat{r}$ is the estimated phase count. True phase counts are 1 for scalar signals, 2 for two waves and 3 for the curved torus.

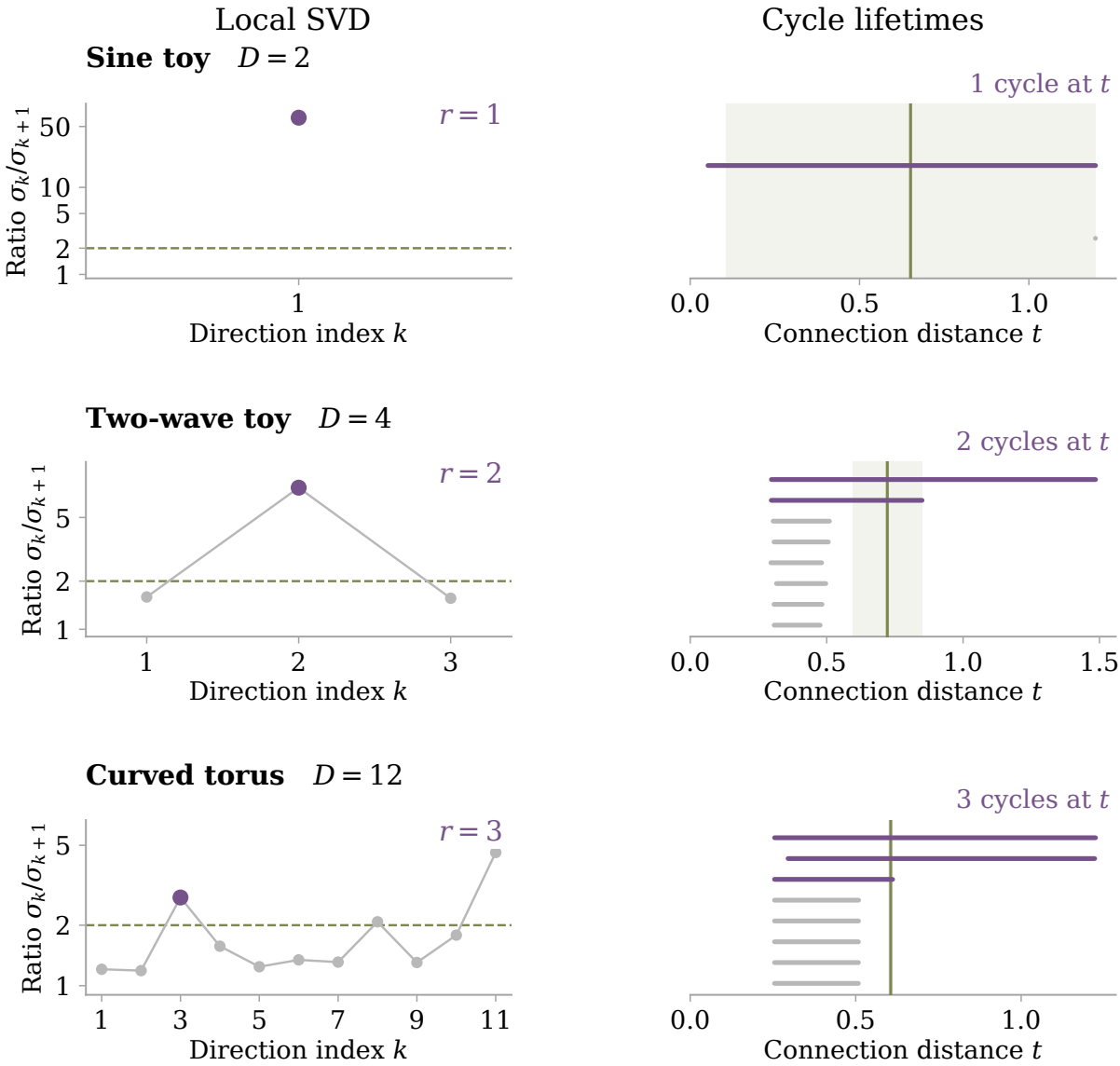


Figure 5: Choosing r in three noise-free examples. Left: the first median local singular-value ratio $\sigma_k/\sigma_{k+1} \geq 2$ proposes r (purple). Right: bars show cycle lifetimes; the chosen t (olive) crosses exactly r bars; shading marks the accepted interval. Up to eight longest lifetimes are displayed; all cycles enter the count. Balls of radius $t/2$ cover all training windows.

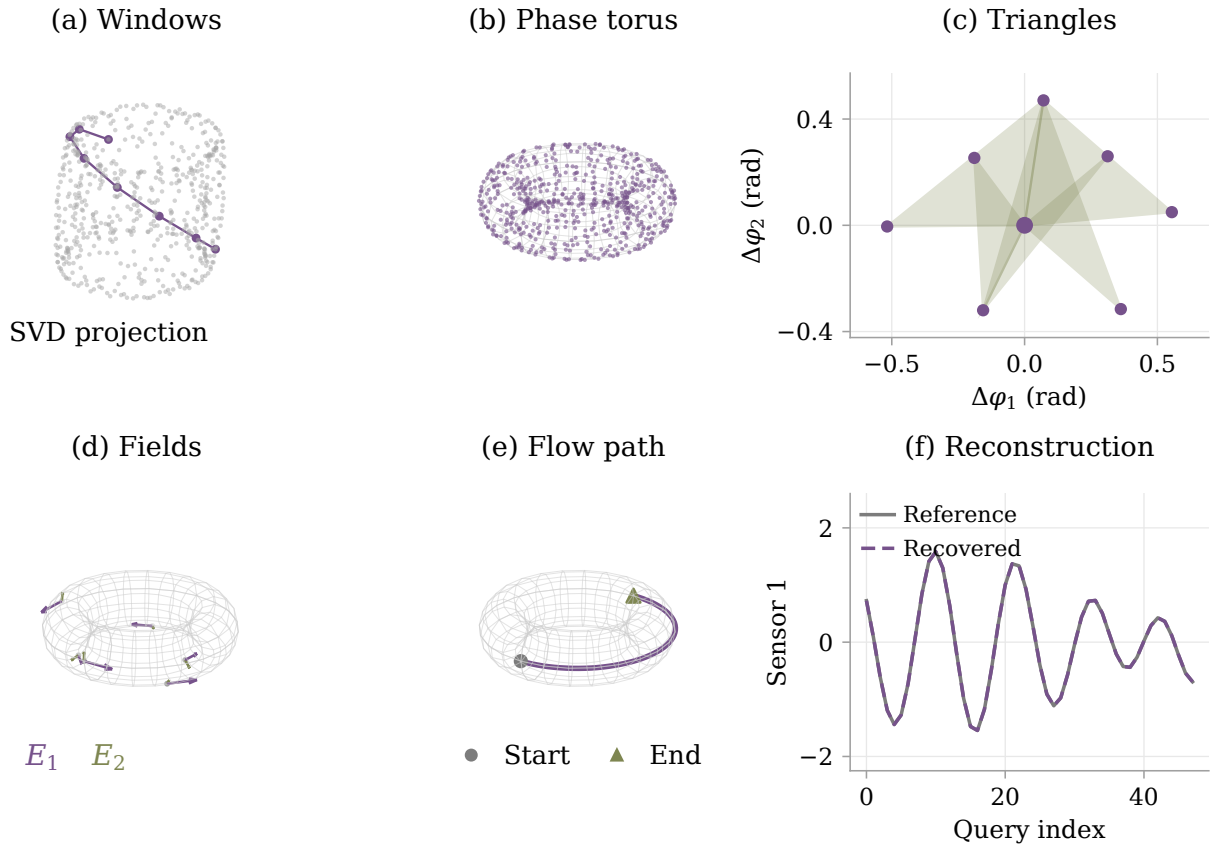


Figure 6: TorusFlow on the clean two-wave toy. (a) Window coordinates from SVD. (b) Recovered phases displayed on a torus. (c) Six triangles from the fitted complex. (d) Estimated phase fields. (e) A path obtained by integrating the first field. (f) Reconstruction at sensor 1. The torus displays phase coordinates; sensor positions are shown in Figure 4.

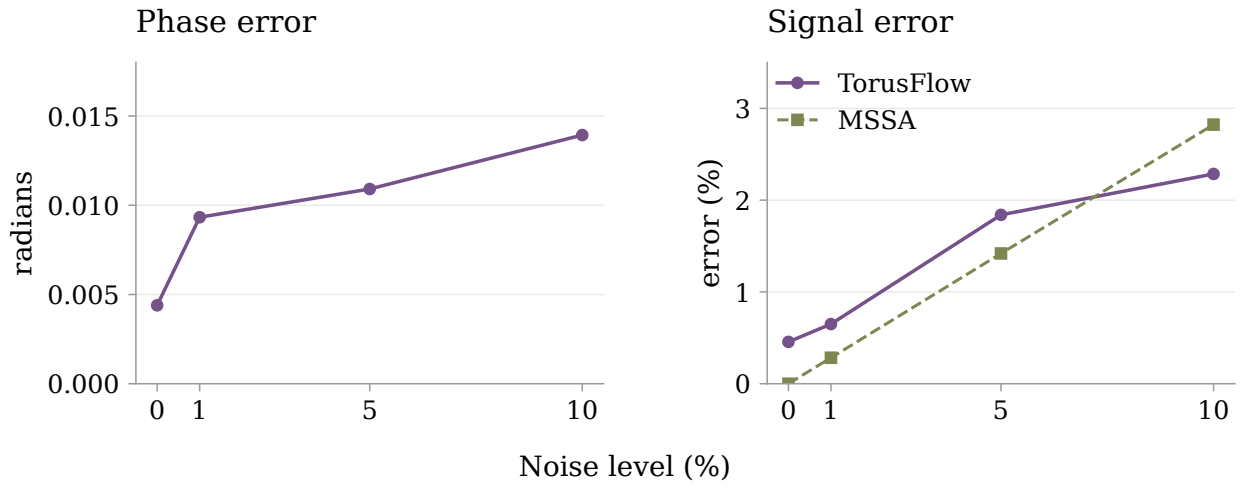


Figure 7: Noise sensitivity on the three-phase torus. Left: phase error of TorusFlow. Right: signal error of TorusFlow and MSSA.

Comparison

SSA/MSSA [1, 2], circular coordinates (CC) [5] and toroidal coordinates (TC) [12] receive the same raw inputs and are evaluated at the same times. The two cases are the nonlinear single-sensor signal with 10% noise and the two-wave signal with 5% noise, including their trends. A +CDF variant rescales its angles using their training empirical distribution. A dash denotes an output that was not evaluated or, for Koopman phases, a failed complete primitive phase-basis check.

	Phase error (rad)		Signal error (%)	
Method	Nonlinear	Two waves	Nonlinear	Two waves
TorusFlow	0.01157	0.00431	2.499	0.695
SSA	–	–	3.753	1.728
MSSA	–	–	3.753	0.702
CC	0.38152	0.11481	–	–
TC	0.38152	0.12429	–	–
CC+CDF	0.00983	0.08566	–	–
TC+CDF	–	0.09174	–	–
Koopman	–	0.06271	2.299	5.430

Table 3: Comparison on the same test inputs and evaluated times. The nonlinear and two-wave cases use 10% and 5% noise, respectively. CDF calibration uses training data only.

			Phase error (rad)		Signal error (%)	
Signal	Noise (%)	$\widehat{r}_{\mathrm{K}}$	TorusFlow	Koopman	TorusFlow	Koopman
Sine toy	0	1	0.00020	0.00778	0.496	0.981
Two-wave toy	0	2	0.00134	0.06535	0.110	5.373
One nonlinear sensor	10	2	0.01157	–	2.499	2.299
Two waves	5	2	0.00431	0.06271	0.695	5.430
Curved torus	0	3	0.00439	–	0.456	3.914
Curved torus	1	3	0.00933	–	0.650	3.549
Curved torus	5	3	0.01091	–	1.840	3.583
Curved torus	10	3	0.01393	–	2.285	3.915

Table 4: Koopman baseline [13] on all eight identical test blocks. $\widehat{r}_{\mathrm{K}}$ is the number of selected Koopman phases. A dash marks an incomplete or nonprimitive phase basis; signal reconstruction uses the full spectral basis and can still be evaluated. Implementation: Appendix A.8.

Appendix A. Implementation details

A.1. Preparation and candidate search

These settings define signal preparation and the search over window lengths and neighbourhood sizes.

Operation	Setting
MSSA	Lag 32; rank at most 24, selected by masked cross-validation. Mask every delay copy of each held-out measurement.
Trend filter	Savitzky–Golay window 129, degree 2.
Boundary trim	95 samples at each end. Keep original timestamps and reindex the retained block of length N_{p}.
Channel scaling	Replace scales below 10^{-12} times the largest by 1; reject an entirely constant record.
SVD	Relative numerical cutoff 10^{-10}. Reduce at the largest consecutive singular-value ratio of at least 10.
Reduction check	Keep the numerical span if the discarded relative norm exceeds 0.05 or a resolved local-dimension estimate changes.
Window candidates	$L \in \{4, 8, 16, 32, 48\}$, subject to $mL \le 128$.
Neighbour candidates	$k \in \{24, 48, 96\}$.
Field landmarks	At most 64, chosen by farthest-point sampling starting at the first training window.

Retain the first candidate passing all training checks. Reduction checks cannot guarantee preservation of arbitrarily weak phase directions.

Let $\mu \in \mathbb{R}^m$ contain training residual means and let diagonal S contain channel scales. Prepared measurements and windows are

$$y_n^{\mathrm{p}} = \frac{S^{-1}(y_n^{\mathrm{osc}} - \mu)}{\sqrt{mL}}, \qquad x_n^{\mathrm{p}} = ((y_n^{\mathrm{p}})^\top, \dots, (y_{n+L-1}^{\mathrm{p}})^\top)^\top, \qquad K_{\mathrm{p}} = N_{\mathrm{p}} - L + 1.$$

Preparation is assumed to preserve the observable phase directions. Its noiseless observation map g^{p} defines the delay map h_L^{p} and $M = h_L^{\mathrm{p}}(\mathbb{T}^r)$. Geometry queries at $z \in \mathbb{R}^{mL}$ use $\widetilde{z} = U_D^\top z \in \mathbb{R}^D$.

A.2. Tangent planes

Estimate the r tangent directions by differentiating fitted surface equations.

Quadratic fit. Use a feature vector $f(\widetilde{z})$ containing 1, all coordinates and all pairwise products. Fit relations

$$g_a(\widetilde{z}) = \widetilde{z}^\top C_a \widetilde{z} + a_a^\top \widetilde{z} + b_a, \qquad C_a = C_a^\top \in \mathbb{R}^{D\times D}, \quad a_a \in \mathbb{R}^D,\ b_a \in \mathbb{R}.$$

Normalize each relation's coefficient vector to unit norm. Set $G(\widetilde{z})_{a,:} = \nabla g_a(\widetilde{z})^\top$ and let $\lambda_1 \le \cdots \le \lambda_D$ be the eigenvalues of $G^\top G$. Exact relations give the tangent plane $\ker G$; approximate relations use its r smallest right singular directions.

Check	Rule
Feature budget	$1 + D + D(D+1)/2 \le \min(K_{\mathrm{p}}/2, 1024)$.
Fit/validation split	Fit four fifths of TRAIN; validate on every fifth row.
Row residual	$\max_a \lvert g_a(\widetilde{z})\rvert / \lVert f(\widetilde{z})\rVert_2$; use the maximum over rows.
Exact fit, tried first	Relative tolerance 10^{-10} and gradient rank $D - r$ throughout TRAIN, with eigenvalue tolerance 10^{-8}.
Approximate fit	Reuse the feature SVD with cutoffs $\tau = 10^{-4}2^j$, $j = 0, \dots, 6$; require fitting and validation residuals at most 3τ.

For an approximate fit, require throughout TRAIN

$$\frac{\lambda_{r+1}}{\lambda_D} \ge 10^{-8}, \qquad 1 - \frac{\lambda_r}{\lambda_{r+1}} \ge 0.1.$$

Choose the cutoff maximizing the worst validation value of $(\lambda_{r+1} - \lambda_r)/\lambda_D$; break ties in favour of the smaller cutoff. Check spectral separation and data support at each query. If the quadratic fit fails, use local cubic surfaces.

Local cubic fallback. At every retained window $a_i = \widetilde{x}_i$, fit a polynomial surface:

1. Set $q_g = \binom{r+3}{3} - 1$ and use the $k_g = \min\{K_{\mathrm{p}} - 1, \max(k, q_g + 2)\}$ nearest other windows. Reject if $k_g < q_g$. Write their indices as $\mathcal{N}_i$ and ordered distances as $d_{i,j}$.
2. Let h_g be the median positive $d_{i,\min(8,k_g)}$ and set $\rho_i = \max\{d_{i,k_g}, 10^{-6} h_g\}$. Reject if h_g is undefined. PCA of $a_j - a_i$ gives $\widehat{Q}_i \in \mathbb{R}^{D\times r}$.
3. For $\delta_{ij} = (a_j - a_i)/\rho_i$ and $u_{ij} = \widehat{Q}_i^\top \delta_{ij}$, fit a normal polynomial $p_i : \mathbb{R}^r \to \mathbb{R}^D$ of degrees 1 through 3:

$$\min_{p_i} \sum_{j\in\mathcal{N}_i} e^{-\|\delta_{ij}\|^2} \left\| p_i(u_{ij}) - (I_D - \widehat{Q}_i\widehat{Q}_i^\top)\delta_{ij} \right\|^2, \qquad \widehat{Q}_i^\top p_i \equiv 0.$$

   Use least-squares cutoff 10^{-10}. Twice replace $\widehat{Q}_i$ by an orthonormal basis of $\widehat{Q}_i + Dp_i(0)$ and refit p_i.

At a query $\widetilde{z}$, evaluate

$$g_i(u) = a_i + \rho_i\big(\widehat{Q}_i u + p_i(u)\big),$$
$$u_i(\widetilde{z}) = \widehat{Q}_i^\top(\widetilde{z} - a_i)/\rho_i, \qquad J_i(\widetilde{z}) = \widehat{Q}_i + Dp_i(u_i(\widetilde{z})).$$

For $d_i = \|\widetilde{z} - a_i\| < 4h_g$, normalize

$$w_i(\widetilde{z}) \propto \exp\left[-\frac{d_i^2}{2h_g^2} - \frac{d_i^2}{16h_g^2 - d_i^2}\right].$$

Set $w_i = 0$ outside this support and reject unsupported queries. The leading r eigendirections of

$$\sum_i w_i(\widetilde{z}) J_i(\widetilde{z})\big(J_i(\widetilde{z})^\top J_i(\widetilde{z})\big)^{-1} J_i(\widetilde{z})^\top$$

define $\widetilde{Q}$ and $\widetilde{P} = \widetilde{Q}\widetilde{Q}^\top$. Require an eigenvalue gap greater than 0.1 after direction r. Differentiate the polynomials, weights and separated spectral projector analytically. The same charts give the position estimate $\bar{g}(\widetilde{z}) = \sum_i w_i(\widetilde{z}) g_i(u_i(\widetilde{z}))$.

A.3. Transported fields

Define the field between windows and fit its coefficients from overlap transport.

Compute overlap transport from

$$(J_n^\top J_n) A_n = J_n^\top P_1 Q_n, \qquad A_n \in \mathbb{R}^{r\times r}, \qquad J_n = P_0 Q_{n+1}.$$

At J field landmarks $a_\ell \in \mathbb{R}^D$, use tangent bases $\widetilde{Q}_\ell$ and local affine models

$$u_\ell(\widetilde{z}) = \widetilde{Q}_\ell^\top(\widetilde{z} - a_\ell)/h \in \mathbb{R}^r,$$
$$V_\ell(\widetilde{z}) = \widetilde{Q}_\ell \left(M_{\ell 0} + \sum_{j=1}^{r} u_{\ell,j}(\widetilde{z}) M_{\ell j} \right) \in \mathbb{R}^{D\times r}.$$

The unknown matrices $M_{\ell j} \in \mathbb{R}^{r\times r}$ specify values ($j = 0$) and linear variations ($j \geq 1$). Set $h > 0$ to the median TRAIN distance to the fourth nearest field landmark (or the last available if $J < 4$). Blend the models with normalized Gaussian weights [9]:

$$\eta_\ell(\widetilde{z}) \propto \exp\left(-\frac{\|\widetilde{z} - a_\ell\|^2}{2h^2}\right), \qquad \sum_{\ell=1}^{J} \eta_\ell(\widetilde{z}) = 1.$$

With $\widetilde{P} = \widetilde{Q}\widetilde{Q}^\top$, project onto the query tangent plane:

$$\widetilde{W}(\widetilde{z}) = \widetilde{P}(\widetilde{z}) \left[\mathsf{A}_0 + \sum_{k=1}^{D} \widetilde{z}_k \mathsf{A}_k + \sum_{\ell=1}^{J} \eta_\ell(\widetilde{z}) V_\ell(\widetilde{z}) \right],$$
$$W(z) = U_D \widetilde{W}(U_D^\top z) \in \mathbb{R}^{mL\times r}.$$

Include the global affine block $\mathsf{A}_k \in \mathbb{R}^{D\times r}$ only if $D(D+1) \leq Jr(r+1)$; otherwise set it to zero. Freeze geometry and weights before fitting. At TRAIN windows, set

$$\widetilde{W}_n = \widetilde{W}(\widetilde{x}_n), \qquad R_n = \widetilde{Q}(\widetilde{x}_n)^\top \widetilde{W}_n.$$

Fit all coefficients jointly:

$$\min_{M,\mathsf{A}} \quad \frac{\sum_{n=0}^{K_\mathrm{p}-2} \|R_{n+1} - A_n R_n\|_F^2}{(K_\mathrm{p}-1)r} + \lambda_s \frac{\sum_{n=0}^{K_\mathrm{p}-1} \|\widetilde{W}_n - \widetilde{W}_{\nu(n)}\|_F^2}{K_\mathrm{p} D} + \frac{\lambda_r}{K_f}\left(\sum_{\ell=1}^{J}\sum_{j=0}^{r} \|M_{\ell j}\|_F^2 + \sum_{k=0}^{D} \|\mathsf{A}_k\|_F^2\right), \qquad R_0 = I_r.$$

Here $\nu(n)$ is the nearest other training window, K_f counts included dictionary columns, and $\lambda_s = 10^{-3}$, $\lambda_r = 10^{-6}$. The three terms enforce transport, agreement at nearby states and bounded coefficients.

A.4. Topology and paths

Select a distance scale that resolves r cycles and covers the training windows, then construct labelled paths for the period integrals.

Use Ripser for persistence [16] and DREiMac for cocycles and landmarks [15].

Cycle count. At distance threshold t, let N_v, N_e, N_c be the numbers of vertices, edges and connected components. The triangle-boundary matrix ∂_2 has edges as rows and triangles as columns, with entries ± 1 along each oriented boundary and zero elsewhere. Then

$$b_1(t) = N_e - N_v + N_c - \operatorname{rank}_{\mathbb{F}_{47}} \partial_2.$$

The graph contributes $N_e - N_v + N_c$ independent cycles; triangle boundaries remove the rank term. Here $\mathbb{F}_{47}$ denotes arithmetic modulo 47. Ripser tracks these cycles across thresholds.

Operation	Setting
Initial complex	At most 192 topology landmarks.
Refinement	384, 768, 1024, 1536, then at most 2048 landmarks. Keep the selected resolution for the period complex.
Final filtration cap	1.1 times the previous common death; if unavailable, use the RMS norm of $\widetilde{x}_n$. Use the same cap for periods.
Local dimension estimate	At 64 uniformly selected TRAIN windows, use 24 neighbours. The first median consecutive singular-value ratio at least 2 proposes r.
Cocycle arithmetic	Modulo 47, followed by an integer lift.
Path graph	24 nearest neighbours, restricted to cover-supported segments.
Segment integrals	Three-point Gaussian quadrature, in both directions.

Choose the dimension and cover. Order persistent classes by observed lifetime. For a class surviving the maximum tested distance (the filtration cap), its death and lifetime have only lower bounds. The first r classes must share an interval with $2b < d$, where b is the latest birth and d the earliest death. If local geometry supplies no dimension, choose the prefix maximizing the lifetime ratio of the last included class to the first excluded class, requiring a ratio of at least 2. Do not infer a dimension from lifetime ratios involving cycles still present at that cutoff alone. Require $1 \le r < D$. This dimension rule is a finite-data heuristic.

Let ϵ_{land} be the largest distance from TRAIN to its nearest landmark, and let δ be half the median positive nearest-neighbour distance within TRAIN. Split

$$\max\{2b,\, 2(\epsilon_{\mathrm{land}} + \delta)\} < t < d$$

at persistence births and deaths. Choose the midpoint of the widest interval with exactly r active classes, strictly below any filtration cap, and set $a_{\mathrm{cov}} = t/2$. Use this rule both before field fitting and for the period complex. The resulting cover contains the δ-neighbourhood of TRAIN; arbitrary new observations may lie outside it.

Check the integer cycles. Lift cocycle labels modulo 47 to integers while preserving triangle closure; apply signed integer corrections when needed. Require:

- a connected complex covering all training windows;
- first cohomology dimension exactly r modulo 47;
- column Hermite normal form I_r for label sums around the fundamental cycles of a spanning tree.

This certificate applies to the finite complex.

Construct integration paths. Join topology landmarks by shortest paths on the supported graph. For quadratic geometry, use verified chords. For local cubic geometry, use the polynomial path

$$\gamma_{uv}(s) = (1-s)g_u\Big(s\widehat{Q}_u^\top \Delta/\rho_u\Big) + s\, g_v\Big((s-1)\widehat{Q}_v^\top \Delta/\rho_v\Big), \quad \Delta = a_v - a_u, \quad 0 \le s \le 1,$$

only if its control polygon lies within one covering ball; otherwise use a verified chord. Integrate along the fine edges of the selected paths in both directions. Require their label sums to equal the corresponding topology-edge labels.

A.5. Phase interpolation

These weights extend the landmark phase estimates to nearby windows while preserving the value and derivative at each landmark. For $d_v = \|U_D^\top(z - z_v)\|$, blend local phase estimates with

$$\rho_v(z) \propto \begin{cases} d_v^{-4} \exp\!\left(-\dfrac{d_v^2}{a_{\mathrm{cov}}^2 - d_v^2}\right), & 0 < d_v < a_{\mathrm{cov}}, \\ 0, & d_v \ge a_{\mathrm{cov}}, \end{cases} \qquad \sum_v \rho_v(z) = 1.$$

Only landmarks within a_{cov} contribute; nearer landmarks receive larger weights. At $z = z_v$, set $\rho_v = 1$ and all other weights to zero by continuity. The resulting phase map matches the local estimate and its derivative at that landmark.

A.6. Acceptance and reconstruction

The thresholds determine whether a fitted model is accepted. The integration settings specify how its fields reconstruct windows.

Normalize training flow errors by $\sigma_x = (K_{\mathrm{p}}^{-1} \sum_n \|x_n^{\mathrm{p}}\|^2)^{1/2}$. Inverse errors include directions discarded by SVD.

Diagnostic	Threshold
Relative edge-integral correction	0.05
Normalized RMS error in $D\Phi\, E - I_r$	0.15
RMS phase-increment error (radians)	0.05
Relative inverse window error	0.05
Relative full-turn return defect	0.05
Relative forward/reverse flow-order discrepancy	0.05
Relative new-window projection error	0.10
Relative overlap mismatch before averaging	0.10

Also require $\sigma_{\min}(J_n) \ge 10^{-3}$, $\mathrm{cond}(\Pi) \le 10^8$, full field rank, tangent spectral separation and coverage. Test inverse reconstruction and flow order at 16 uniformly selected TRAIN windows; test full turns from z_*. Use RK4, phase representatives in $(-\pi, \pi]$, a fixed flow order and angular steps at most 0.12. Reject a trajectory leaving geometric support.

Optional normal correction. With local cubic geometry, if ambient inverse reconstruction or the full-turn test fails, retry with a normal correction after every RK4 step:

$$\widetilde{z} \leftarrow \widetilde{z} + \big(I_D - \widetilde{Q}(\widetilde{z})\widetilde{Q}(\widetilde{z})^\top\big)\big(\bar{g}(\widetilde{z}) - \widetilde{z}\big).$$

Here $\bar{g}$ is the position estimate from the frozen local charts. This is a geometry-constrained finite inverse; it changes the ambient ODE.

Correction check	Rule
Normal residual	Below $10^{-9}\sigma_x$ within 20 iterations; otherwise reject.
Reference point	Correct z_* and reset the phase origin there; require displacement below $0.05\sigma_x$.
Full turns	Test from 8 training windows.
Step selection	Largest step in $\{0.12, 0.06, 0.03\}$ passing the original checks and giving a return-point difference below $0.005\sigma_x$ when halved.
Constrained phase speed	Central differences with angular increments $\pm 10^{-4}$ at 16 training windows.

New observations. Repeat coverage, duality, reconstruction and overlap checks. Compare phase increments with the training increment and normalize inverse errors by the unscaled residual-window norm. Nonfinite diagnostics, failed checks or exhausted search leave reconstruction unresolved. These tests certify neither global topology nor convergence.

A.7. Evaluation protocol

The table gives the training and test splits, evaluated samples and window lengths.

Example	TRAIN/test	Scored samples	Windows	L
Sine ($\Delta t = 0.4$)	1200/400	210	207	4
Nonlinear scalar	1800/320	130	123	8
Two waves (clean and 5% noise)	2400/320	130	127	4
Curved torus (0, 10% noise)	2400/320	130	123	8
Curved torus (1, 5% noise)	2400/320	130	127	4

Phase alignment uses TRAIN only: an integer character match and constant phase offsets. All reported valid matches in these runs are signed permutations; incomplete or nonprimitive Koopman matches remain unresolved. Reconstruction conditions on the observed test block. SSA/MSSA controls use lag 32; CC/TC use our selected lag, MSSA conditioning and trend filter. Saved controls are reused after verifying identical raw inputs, preprocessing and evaluated times.

A.8. Koopman baseline

This baseline estimates oscillations from a spectral model of how windows evolve.

We follow Sections 4.2–4.3 and 5, and Algorithms 1–2 of Giannakis [13]. Algorithm 1 estimates sampling density with eight nearest neighbors and selects kernel bandwidths at the maximum logarithmic kernel-sum slope over $\epsilon = 2^l$, $l = -30, -29.9, \ldots, 10$. For large records, the bandwidth scan uses a fixed sample of 32768 off-diagonal pairs plus the exact diagonal contribution; the final kernel and eigensystem use every training window. The variable-bandwidth diffusion operator supplies basis functions ϕ_j, stationary weights w and normalized Laplace eigenvalues $\eta_j = \log \kappa_j / \log \kappa_1$. We use the reversible normalization $w_i \propto d_i/q_i$ from Equation (24).

In this weighted L^2 basis, the eigenproblem is

$$\big(V - \varepsilon \operatorname{diag}(\eta)\big)a = \gamma a, \qquad V_{ij} = \sum_n w_n \phi_i(x_n) \frac{\phi_j(x_{n+1}) - \phi_j(x_{n-1})}{2\Delta t}.$$

This is the equivalent form of the Sobolev-basis problem in Equations (21), (32) and (33). Endpoints are omitted. We replace V by $(V - V^\top)/2$ to preserve skew-adjointness at finite sample size and keep the constant as an exact null mode. Eigenfunctions are ordered by their normalized Dirichlet energy.

We select 32, 64 or 128 basis functions and $\varepsilon \in \{10^{-4}, 10^{-3}, 10^{-2}\}$ on the original TRAIN block. The first 75% of its prepared windows fit the candidate model; the last 25%, after a gap of $L - 1$ window starts, validate it. The loss is the equally weighted mean of window reconstruction MSE and one-step harmonic prediction MSE. The shared preprocessing uses the original TRAIN block; no test observations enter selection. The selected model is refit on all training windows. A weighted least-squares Gram solve represents the observation windows in the full regularized eigenfunction basis. Artificial diffusion decay is omitted when advancing the harmonic modes for validation.

The integer-dependence check uses coefficients bounded by 4 and a frequency resolution $\pi/((K-1)\Delta t)$. Because central differences distort frequencies at finite sampling intervals, we refine frequencies from the circular mean of successive training eigenfunction phase increments, using the sampled evolution identity in Equation (5). Dependence also requires training phase-product coherence of at least 0.9, avoiding a rejection based only on an accidental close frequency relation. This is explicit finite-sample postprocessing of Algorithm 2. Candidates with modulus coefficient of variation above 0.35 or relative central-derivative residual above 0.5 are excluded. The rounded kernel dimension estimate supplies a phase-count cap between one and four. These finite choices do not certify a complete primitive phase lattice. After freezing predictions, TRAIN reference phases determine an integer gauge and origins for evaluation. Phase error is reported only when the selected count agrees with the reference and the fitted gauge has determinant ± 1; all failures remain in the saved records.

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